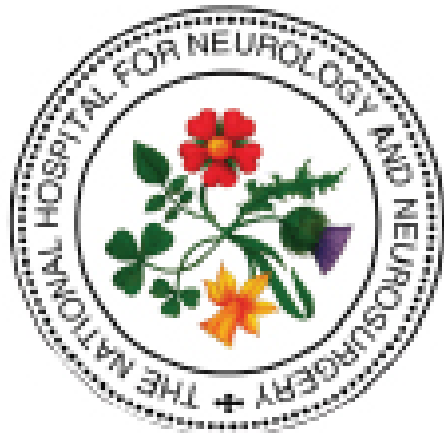
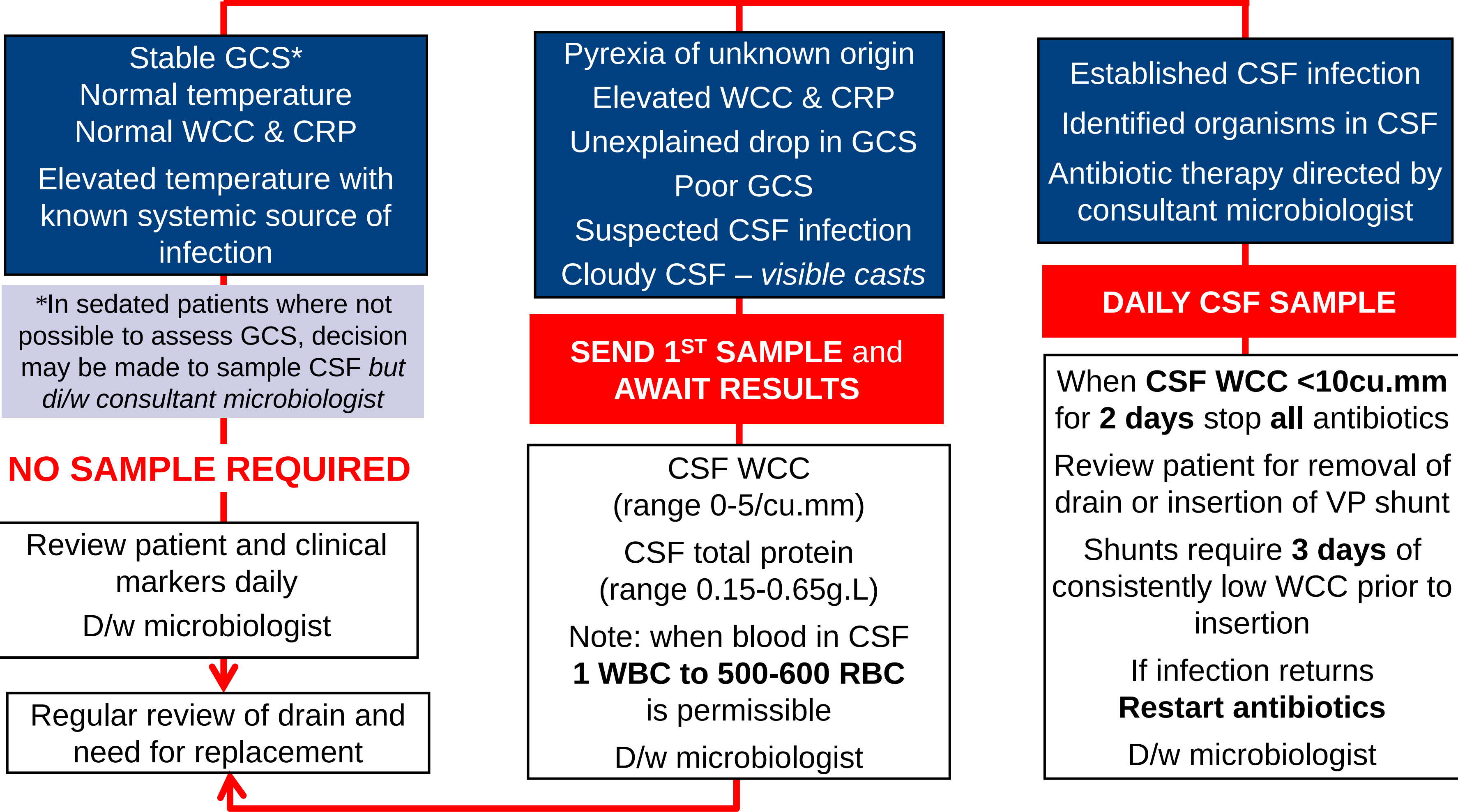




The National Hospital for Neurology and Neurosurgery
Neuroscience Critical Care Unit (NCCU)
EXTERNAL VENTRICULAR DRAIN (EVD)
INDICATORS for CSF SAMPLING

DOES THE PATIENT HAVE ANY OF THE FOLLOWING SIGNS OR SYMPTOMS?



TREATMENT of VENTRICULITIS	
BACTERIAL VENTRICULITIS - Antibiotic therapy required - In patients with VP or VA shunt this requires removal of infected system and insertion of EVD - Antibiotics are given intraventricularly and systemically - Treatment is modified according to antibiotic sensitivities - Treatment is continued until infection is eradicated and CSF white cell counts are consistently low at which time a formal procedure (shunting or removal of EVD) can be planned	
EMPIRICAL and TARGETED TREATMENT	
<i>Treatment should be modified in light of antimicrobial sensitivities of isolate</i>	
Empirical treatment (organism unknown or likely mixed infection)	Gentamicin 5mg/1ml intraventricular 24 hourly Plus Vancomycin 20mg/4ml intraventricular 24 hourly
Gram positive infections (Coagulase-negative Staphylococci, <i>S.aureus</i> , <i>Propionibacterium spp</i>)	Vancomycin 20mg/4ml intraventricular 24 hourly
Gram negative infections (<i>Klebsiella spp</i> , <i>E.coli</i>)	Gentamicin 5mg/1ml intraventricular 24 hourly
Resistant Gram negative infections (<i>P.aeruginosa</i> , <i>Enterobacter spp</i> , <i>Klebsiella spp</i> resistant to cephalosporins, <i>but usually sensitive to carbapenems</i>)	Amikacin 10mg/4ml intraventricular 24 hourly
Fungal infections	Amphotericin B 0.025mg/4ml intraventricular <i>as test dose</i> <i>Subsequent doses discussed with Microbiologist</i>

ANTIBIOTICS for INTRAVENTRICULAR USE			
<i>* Commonly used</i>			
Drug	Standard dose used at NHNN	Flush	Total volume
*Gentamicin	5mg in 1ml	4ml sterile saline	5ml
*Vancomycin	20mg in 4ml	4ml sterile saline	8ml
Amikacin	10mg in 4ml	4ml sterile saline	8ml
Amphotericin B	0.025mg in 4ml as test dose <i>Subsequent doses to be discussed with Microbiology</i>	4ml water for injection	8ml
<i>NHNN do not routinely check vancomycin or gentamicin levels for adults Patients are observed for toxicity (neuro-irritant effect) and effectiveness (CSF, WBC, CSF cultures)</i>			

SYSTEMIC ANTIBIOTICS COMMONLY USED IN CONJUNCTION WITH INTRAVENTRICULAR ANTIBIOTICS	
GRAM POSITIVE INFECTIONS	LINEZOLID 600 mg PO / IV 12 hourly (D/w Microbiologist)
GRAM NEGATIVE INFECTIONS	CEFTAZIDIME 2g IV 8 hourly (Or other agent as d/w Microbiologist)
MULTI-RESISTANT GRAM NEGATIVE INFECTIONS	MEROPENEM 2g IV 8 hourly (D/w Microbiologist)
FUNGAL INFECTIONS	VORICONAZOLE (D/w Microbiology)